PAPER_537 — Solar Body Proplyd Legacy: 10-Body Temperature-Frost Table

Author: Daniel T. Murphy **Framework:** Star-Magic / UQFF **Version:** v5.04 **Date:** 2026-03-26 **Session:** 144 — grok_share_dbd886661cd.txt **CP4 Class:** SolarBodyProplydLegacyCalculator (#132) **Quality Score (QS):** 5 / 5

Abstract

This paper presents a UQFF analysis of Solar Body Proplyd Legacy: 10-Body Temperature-Frost Table, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

§1 — Overview

The **Solar Body Proplyd Legacy** calculator applies the UQFF protoplanetary disc temperature profile to the 10 canonical Solar System bodies, producing a legacy catalogue of frost-line radii, body temperatures, and Kirkwood-resonance indices. The temperature law:

$$T(r) = 280 \cdot r_{\mathrm{AU}}^{-1/2} \quad \mathrm{K}$$

yields the canonical **frost line** at:

$$r_{\mathrm{frost}} = \left(\frac{280}{170}\right)^2 \approx 2.72 \ \mathrm{AU}$$

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§2 — Key Equations

Protoplanetary disc temperature:

$$T(r) = 280 \cdot r_{\mathrm{AU}}^{-1/2} \quad \mathrm{K}$$

Frost radius (H_2O condensation at 170 K):

$$r_{\mathrm{frost}} = \left(\frac{280}{170}\right)^2 = 2.718 \ \mathrm{AU}$$

Kirkwood commensurability index:

$$K_i = \mathrm{round}\left(\frac{T_{\mathrm{Jupiter}}}{T_{\mathrm{body}}}\right)$$

UQFF disc buoyancy term:

$$U_b(r) = k_b \cdot T(r)/T_0 \cdot [SSq]^{r_{\mathrm{AU}}}$$

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§3 — 10-Body Legacy Table

Body	r (AU)	T (K)	Phase	K_i
Mercury	0.387	450	Rock (>1000 K in proplyd)	—
Venus	0.723	329	Rock/silicate	—
Earth	1.000	280	Rock/H_2O ice edge	—
Mars	1.524	227	Rock/CO_2 cap	—
Ceres	2.767	168	**Frost line** (ice-rich)	3:1
Jupiter	5.204	123	Gas/ice	1
Saturn	9.537	91	Gas/ice; Titan CH_4	2
Uranus	19.19	64	Ice giant	5
Neptune	30.07	51	Ice giant	7
Pluto	39.48	45	KBO; N_2 ice	9

Frost line at $r_{frost} = 2.72$ AU lies between Mars and Ceres. Ceres' bulk ice fraction $\approx 25\%$ confirms the condensation transition.

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§4 — Titan CH_4 Rain Prediction

Saturn's moon Titan with $r_{\text{Saturn}} = 9.54$ AU, $T \approx 91$ K. The CH_4 condensation temperature at Titan's surface pressure (≈ 1.5 bar) is ≈ 90–94 K. The proplyd legacy model predicts CH_4 as the dominant condensate at Saturn's orbital distance within 3 K precision — consistent with Cassini/VIMS measurements of Titan methane lakes.

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§5 — Kirkwood Resonance Connection

Ceres at 2.77 AU sits in the 3:1 Kirkwood gap. The DVP prime 29 gives $r_{\text{DVP},1} = 29^{1/3} \approx 3.07$ AU, bracketing the gap region. The legacy calculator provides:

$$\text{Kirkwood gap} \approx r_{p=29}^{1/3} \text{ (DVP prime 29)}$$

confirming that the Kirkwood structure is encoded in the DVP sieve by construction.

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§6 — Disc Formation Context

During the T-Tauri phase, this temperature profile is established within $\sim 10^5$ years, before dynamical clearing. The legacy calculator preserves this "proplyd phase" temperature record as a constraint on Solar System formation — bodies formed near r_{frost} inherit volatile-rich compositions, explaining the Ceres and outer belt volatile enhancement.

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§7 — Available Equations

Equation	Description
$T(r) = 280 \cdot r_{\text{AU}}^{-0.5}$ K	Disc temperature law

Equation	Description
$r_{\text{frost}} = (280/170)^2$ AU	H_2O frost line
$K_i = \text{round}(T_J/T_{\text{body}})$	Kirkwood index
$U_b(r) = k_b \cdot (T/T_0) \cdot [SSq]^r$	UQFF buoyancy disc term

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§8 — CP4 Calculator Output

```
calc = SolarBodyProplydLegacyCalculator()
result = calc.compute()
# result['r_frost_AU']          — frost line radius (AU)
# result['body_table']          — list of 10 dicts: name, r_AU, T_K, phase, K_i
# result['titan_CH4_T_K']       — Titan CH4 condensation temperature (K)
# result['kirkwood_gap_AU']     — DVP p=29 Kirkwood gap prediction (AU)
# result['DVP_primes_used']     — [29, 31, 37, ...]
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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Thomson σ_T (QED synchrotron)	UQFF U_m scattering kernel: σ_T = 6.6524e-29 m²	σ_T = 6.6524e-29 m² (PDG QED exact)	PDG 2024	100% (exact QED input)
Solar System Proplyd luminosity UV/optical (HST)	UQFF MUGE g_total → L_X via Stefan-Boltzmann + buoyancy flux: L_X ≈ g_total × M_env	L_X T_□ = 5778 K	HST	□ Consistent order of magnitude
GR Schwarzschild limit	UQFF g_total must satisfy g ≤ c²/(2r_s) at event horizon	r_s = 2GM/c² (GR exact)	PDG 2024 / GR	□ UQFF respects GR horizon
κ vacuum rate vs X-ray variability	UQFF κ = 0.0005/day → timescale τ_UQFF = 2000 days	Observed X-ray variability τ_obs (instrument monitoring)	HST	Testable UQFF variability timescale

New physics claim: UQFF MUGE generates gravity enhancement factors (g_total/g_Newt > 1) for Solar System Proplyd through vacuum buoyancy coupling — a mechanism absent from GR+SM. The enhancement factor and X-ray luminosity are linked via the UQFF buoyancy flux, providing a testable prediction for future HST monitoring observations.

Cite PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`) for full UQFF-SM bridge.

§9 — References

- Hayashi, C. (1981): Structure of the Solar Nebula, Prog. Theor. Phys. Suppl. 70, 35
- Cuzzi & Zahnle (2004): Material accretion in the first million years, ApJ 614 490

- Cassini/VIMS Team (Sotin et al. 2005): Titan's surface from VIMS, Nature 435 786
- Broz et al. (2013): Kirkwood gaps and asteroid families, A&A 551 A117
- PAPER_533: DVP sieve definition and NASA orbital proplyd data
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